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NPS/003/006 – Technical Specification for Primary 11kv and 20kv Circuit Breakers

1. Purpose

This document is the technical specification for primary 11kV and 20kV circuit breakers to be utilised on the NEDL and YEDL distribution networks.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NEMS 1	Issue 2	Indoor 11kV and 20kV single busbar metalclad extensible primary switchgear and ancillary equipment
	Rev.10	Bulk Call-Off Contract: 11kv Primary Substation 1250 & 2000A Switchboards: Technical Specification
DC/PRC/S/201 Part 1		11 kV Circuit Breakers for Installation at Primary Substations and Supply Points

2. Scope

This specification details the requirements for primary 11kV and 20kV switchgear to be utilised on the NEDL and YEDL distribution networks. It will be necessary to consider and include any project specific requirements as detailed in Appendix 3, Addendum To Supplier Requirements.

3. Technical Requirements

All switchgear shall comply fully with the requirements of the latest version of Energy Networks Association Technical Specification (ENATS) 41-36 and the latest versions of standards and regulations to which ENATS 41-36 refers.

The circuit breakers will preferably have been assessed by the ENA Switchgear Assessment Panel (SAP) and have been awarded an ENA Notice of Conformity.

Appendix 1 of this document details the requirements selected within ENATS 41-36, or where the requirements of ENATS 41-36 have been supplemented. Note particularly the specified fault-break duties and X/R ratios.

Appendix 2 shall be completed for each type of circuit breaker unit.

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Appendix 3 contains an Addendum to Supplier Requirements which lists project specific requirements.

4. References

4.1 External Documentation

Reference	Title
ENATS 12-11	Indoor and outdoor cable boxes for switchgear for service at system voltages of 6.6, 11 and 33kV.
ENATS 41-36	Distribution Switchgear for Service up to 36kV (Cable and Overhead Conductor Connected)

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
K.Midgley	Standards Engineer	K.Midgley	01/02/2006

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G.Bolden	Purchasing Manager	G.Bolden	09/02/2006

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D.Gazda	Standards Team Co-ordinator	D.Gazda	09/02/2006

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S.Johnson	24/03/2006

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	13/02/2006

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	20/02/2006

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APPENDIX 1: Options Selected within ENATS 41-36

11kV circuit breaker

	ENATS 41-36 Ref	
Particular of System		
Voltage		11kV
Frequency		50 Hz
Number of Phases		3
Neutral Earthing		Solid
Circuit Breaker Characteristics		
Number of poles		3
Class – indoor/outdoor	1.2	Indoor
Rated voltage	1.4.1	12 kV Minimum.
Rated Insulation level. (lightning impulse voltage)	1.4.2	95 kV (110 kV across isolating distance)
Rated Frequency	1.4.3	50 Hz
Rated Normal Current	1.4.4	
a) Busbar		1250A and 2000A
b) Circuit-breaker (TX incomer)		1250A and 2000A
c) Circuit breaker (Bus Section)		1250A and 2000A
d) Circuit Breaker (Feeder)		630A and 800A
Rated Short-time withstand current	1.4.5	20kA or 25kA Min
Rated duration of short circuit	1.4.7	3 sec
Rated supply voltage of opening and closing devices and auxiliary and control circuits.	1.4.8	
a) Closing and tripping		110 V DC
b) Indication		110V DC
c) Control		24V DC (Telecontrol)
Rated supply frequency of closing and auxiliary circuits.	1.4.9	DC
Rated short-circuit breaking current. Equal to rated short-time withstand current.	2.4.101	20 kA @ X/R of 24 (time const 76.4 ms) The methodologies in IEC 62271-100 and IEC 62271-308 give this an equivalent rating of; 25kA @ X/R of 14 (time constant 45ms)
Rated short-circuit making current	2.4.103	50 kA
Rated operating sequence.	2.4.104	O – 0.3s – CO – 15s – CO as required by ENATS 41-36 and also: O – 10s – CO – 10s – CO to comply with YEDL and NEDL auto-reclose sequences
Rated capacitive switching currents : Rated cable-charging current.	2.4.107	25 A (Class C1 sub clause 4.107 of IEC 60056)
Classification of mechanical	2.4.110	M2: 10000 operating cycles

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operations M1 or M2		
Classification of electrical endurance	2.4.111	Class E2 Auto reclose
Mechanism type	2.5.5 –2.5.7	Tenderer to state
Key interlock	2.5.11	1 Key
Gas monitoring	1.5.9	Gauge with 2 stage monitoring: 1st stage – Alarm. 2 nd stage – Operation inhibit + alarm
Voltage presence indicating system	1.5.19.1	Shall be taken from fixed part of circuit breaker rather than cable connectors to allow reliable phase check.
Insulation Medium		Non-oil preferred.
Arc extinguishing medium		Non-oil preferred
Colour of Paint		Grey
Power cable termination		To ENATS 12-11: 630A: 1 x 300 sq mm /phase Triplex XLPE 800A: 2 x 300 sq mm/phase Triplex XLPE 1250A: 2x 630 or 3x 400 sq mm /phase XLPE 2000A: 3x 630 or 4x 400sq mm/ phase XLPE

CT Specifications¹

¹ The CT order should be from the table below (Top: 11kV busbar end. Bottom: 11kV circuit end)

1250A Board

Panel Type	CT Application	Continuous Current	Ratio
Transformer 33/11kv	REF	1375A	1/1250T $V_k > (25R_{ct} + 82)$ $I_e < 4mA @ (6R_{ct} + 33)$ $R_{ct} < 6.6\Omega$
	LVOC/DOC	100%	1500/1A 5VA 5P10/ 5VA Class 1.0, $R_{ct} < 11\Omega$
Bus Section			
630A Prot'n IDMT Feeder	OCEF	120%	600/1A 2.5VA 5P20/ 2.5VA Class 1.0 $R_{ct} < 3.8\Omega$
630A Prot'n IDMT +SEF Feeder.	OCEF +SEF	120%	600/1A 2.5VA 5P20/ 2.5VA Class 1.0 $R_{ct} < 3.8\Omega$
630A Prot'n: IDMT + SEF +Auto Reclose Feeder.	OCEF +SEF	120%	600/1A 2.5VA 5P20/ 2.5VA Class 1.0 $R_{ct} < 3.8\Omega$

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630A Unit Prot'n Feeder; Twin Setting IDMT (inc metering)	Unit Protection	690A	1/500T $V_k > (40R_{ct}+59)$ $I_e < 50mA @ 50V$ and $I_e < 100mA @ V_k$
	OCEF	120%	600/1A 2.5VA 5P20/ 2.5VA Class 1.0 $R_{ct} < 3.8\Omega$
	Metering	120%	600/400/5A 15VA 0.2S (Subject to customer requirements)
800A Unit Prot'n Feeder; Twin Setting IDMT (inc metering)	Unit Protection	880A (both ratios)	1/500/750T $V_k > (40R_{ct}+59)$ for 1/500T $V_k > (27R_{ct}+56)$ for 1/750T $I_e < 50mA @ 50V$ and $I_e < 100mA @ V_k$ for both ratios
	OCEF	120%	750/1A 2.5VA 5P20/ 2.5VA Class 1.0 $R_{ct} < 6.9\Omega$
	Metering	120%	750/400/5A 15VA 0.2S (Subject to customer requirements)
Neutral CTs	Standby E/F	100%	500/1A 2.5VA 5P5 3second rating 800%
	REF	1375A	1/1250T $V_k > (25R_{ct} + 82)$ $I_e < 4mA @ (6R_{ct}+33)$ $R_{ct} < 6.6 \text{ ohm}$
2000A Board			
Panel Type	CT Application	Continuous Current	Ratio
Transformer 33/11kv	REF	2200A	1/1250T $V_k > (25R_{ct} + 82)$ $I_e < 4mA @ (6R_{ct}+33)$ $R_{ct} < 6.6 \text{ ohm}$
	LVOC/DOC	150%	1500/1A 5VA 5P10/ 5VA Class 1.0, $R_{ct} < 11\Omega$
Bus Section			
630A Prot'n IDMT Feeder	OCEF	120%	600/1A 2.5VA 5P20/ 2.5VA Class 1.0 $R_{ct} < 3.8\Omega$
630A Prot'n IDMT +SEF Feeder	OCEF +SEF	120%	600/1A 2.5VA 5P20/ 2.5VA Class 1.0 $R_{ct} < 3.8\Omega$
630A Prot'n IDMT + SEF +Auto Reclose Feeder	OCEF +SEF	120%	600/1A 2.5VA 5P20/ 2.5VA Class 1.0 $R_{ct} < 3.8\Omega$

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630A Unit Prot'n Feeder; Twin Setting IDMT (inc metering)	Unit Protection	690A	1/500T $V_k > (40R_{ct}+59)$ $I_e < 50\text{mA @ } 50\text{V}$ and $I_e < 100\text{mA @ } V_k$
	OCEF	120%	600/1A 2.5VA 5P20/ 2.5VA Class 1.0 $R_{ct} < 3.8\Omega$
	Metering	120%	600/400/5A 15VA 0.2S (Subject to customer requirements)
800A Unit Prot'n Feeder; Twin Setting IDMT (inc metering)	Unit Protection	880A (both ratios)	1/500/750T $V_k > (40R_{ct}+59)$ for 1/500T $V_k > (27R_{ct}+56)$ for 1/750T $I_e < 50\text{mA @ } 50\text{V}$ and $I_e < 100\text{mA @ } V_k$ for both ratios
	OCEF	120%	750/1A 2.5VA 5P20/ 2.5VA Class 1.0 $R_{ct} < 6.9\Omega$
	Metering	120%	750/400/5A 15VA 0.2S (Subject to customer requirements)
Neutral CTs	Standby E/F	120%	500/1A 2.5VA 5P5 3second rating 800%
	REF	2200A	1/1250T $V_k > (25R_{ct} + 82)$ $I_e < 4\text{mA @ } (6R_{ct}+33)$ $R_{ct} < 6.6 \text{ ohm}$

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20kV Circuit Breaker

	ENATS 41-36 Ref	
Particular of System		
Voltage		20 kV
Frequency		50 Hz
Number of Phases		3
Neutral Earthing		Solid
Circuit Breaker Characteristics		
Number of poles		3
Class – indoor/outdoor	1.2	Indoor
Rated voltage	1.4.1	24 kV Min.
Rated Insulation level. (lightning impulse voltage)	1.4.2	125 kV (145 kV across isolating distance)
Rated Frequency	1.4.3	50 Hz
Rated Normal Current	1.4.4	630A, 1250A and 2000A 630A, 1250A and 2000A 630A, 1250A and 2000A 630A and 800A
d) Busbar		
e) Circuit-breaker (TX incomer)		
f) Circuit breaker (Bus Section)		
d) Circuit Breaker (Feeder)		
Rated Short-time withstand current	1.4.5	16kA or 20 kA Min
Rated duration of short circuit	1.4.7	3 sec
Rated supply voltage of opening and closing devices and auxiliary and control circuits.	1.4.8	110 V DC 110 V DC 24V DC (Supervisory)
d) Closing and tripping		
e) Indication		
f) Control		
Rated supply frequency of closing and auxiliary circuits.	1.4.9	DC
Rated short-circuit breaking current. Equal to rated short-time withstand current.	2.4.101	16 kA @ X/R of 24 (time const 76.4 ms) The methodologies in IEC 62271-100 and IEC 62271-308 give this an equivalent rating of; 20kA @ X/R of 14 (time constant 45ms)
Rated short-circuit making current	2.4.103	40kA
Rated operating sequence.	2.4.104	O – 0.3s – CO – 15s – CO as required by ENATS 41-36 and also: O – 10s – CO – 10s – CO to comply with YEDL and NEDL auto-reclose sequences
Rated capacitive switching currents : Rated cable-charging current.	2.4.107	31.5A (Class C1 sub clause 4.107 of IEC 60056)
Classification of mechanical operations M1 or M2	2.4.110	M2: 10000 operating cycles
Classification of electrical endurance	2.4.111	Class E2 Auto reclose

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Mechanism type	2.5.5 –2.5.7	Tenderer to state
Key interlock	2.5.11	1 Key (where key is specified)
Gas monitoring	1.5.9	Gauge with 2 stage monitoring: 1st stage – Alarm. 2nd stage – Operation inhibit + alarm
Voltage presence indicating system	1.5.19.1	Shall be taken from fixed part of circuit breaker rather than cable connectors to allow reliable phase check.
Insulation Medium		Non-oil preferred.
Arc extinguishing medium		Non-oil preferred
Colour of Paint		Dark admiralty grey 632 BS381C preferred
Power cable termination		To ENATS 12-11: 630A: 1 x 300 sq mm /phase Triplex XLPE 800A: 2 x 300 sq mm/phase Triplex XLPE 1250A: 2x 630 or 3x 400 sq mm /phase XLPE 2000A: 3x 630 or 4x 400sq mm/ phase XLPE
CT Specifications		
To be specified on project specific basis		

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Appendix 2: Declaration of Technical Performance to be completed by the manufacturer

	MANUFACTURER			
	CIRCUIT BREAKER TYPE REFERENCE			
		ENATS 41-36 reference where applicable	UNIT	
	CIRCUIT BREAKER CHARACTERISTICS			
1	ENATS Certificate of Conformance			
2	Number of poles			
3	Class – indoor/outdoor	1.2		
	Type reference of vacuum bottles (if applicable)			
	Type reference of interrupter (if applicable)			
4	Rated voltage	1.4.1	kV	
5	Rated Insulation level. (lightning impulse voltage)	1.4.2	kV	
6	Rated Frequency	1.4.3	Hz	
7	Rated Normal Current Busbar Circuit-breaker (TX incomer) Circuit breaker (Bus Section) Circuit Breaker (Feeder)	1.4.4	A	
8	Rated Short-time withstand current	1.4.5	kA	
9	Rated duration of short circuit	1.4.7	S	
10	Rated supply voltage of opening and closing devices and auxiliary and control circuits. Closing and tripping Indication Control	1.4.8	V	
11	Rated supply frequency of closing and auxiliary circuits.	1.4.9	Hz/DC	
12	Rated short-circuit breaking current	2.4.101	kA	
13	DC time constant.		msec	
14	Rated short-circuit making current	2.4.103	kA	
15	Rated operating sequence.	2.4.104		
16	Rated capacitive switching currents : Rated cable charging current.	2.4.107	A	

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17	Classification of mechanical operations M1 or M2	2.4.110		
18	Classification of electrical endurance – tested for autoreclose duty – not tested for autoreclose duty	2.4.111		
19	Mechanism type	2.5.5 –2.5.7		
20	Key interlock	2.5.11		
21	Gas monitoring	1.5.9	Type	
22	Voltage presence indicating system (VPIS) fitted	1.5.19.1	Yes/No	
	VPIS: location of capacitor bushing			
23	Insulation Medium			
24	Arc extinguishing medium			
35	Colour of Paint			
	Power cable connection type			
26	Power cable: max number, size and type of cables/phase Transformer incomer Feeder			
27	Test bushing: Location of connection. Type of connection			
	OPERATING MECHANISM DETAILS			
28	Manufacturer and Type Reference of Operating Mechanism.			
29	Mechanism travel.		mm	
30	Closing Coils Number and type Type reference Power consumption		W	
31	Operating Coil-Open Number and type Type reference Power consumption		W	
32	Operating time – Close		msec	
33	Operating time – Open		msec	
34	Closing spring recharge time (motor driven); if applicable		sec	

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35	Spring charge motor: power consumption ; if applicable		W	
36	Closing actuator current; if applicable.		A	
37	Maximum dynamic floor loading		kN	
	SF6 GAS DETAILS			
38	Design pressure of gas system at 20°C Insulation Interrupters		Barg / MPa	
39	Operating pressure of pressure relief device		Barg / MPa	
40	Normal pressure at 20°C		Barg / MPa	
41	Close lockout pressure		Barg / MPa	
42	Trip (open) lockout pressure		Barg / MPa	
43	Minimum close pressure		Barg / MPa	
44	Minimum trip pressure		Barg / MPa	
45	Quantity of gas (mass and volume) per compartment Busbar Feeder Interrupters Total		Kg	
46	Method of monitoring pressure and temperature compensation			
47	Moisture content at commissioning (max.)		mg/kg	
48	Dew point temperature at commissioning (max.)		Deg C	
49	Maximum permissible acidity level (HF)			
50	Maximum permissible leak rate (<1% per annum)			
51	Mass of alumina/molecular sieve and type reference			
	DIMENSIONS AND WEIGHTS			

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52	Transformer incomer Height Width Depth Mass		mm mm mm kg	
53	Bus Section Height Width Depth Mass		mm mm mm kg	
54	Feeder Height Width Depth Mass		mm mm mm kg	
55	Minimum clearances required to achieve internal arc performance from: Rear Side Ceiling Front		mm mm mm mm	

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APPENDIX-3: Addendum to Supplier Requirements.

Project specific installation and protection requirements will be provided by Primary Engineering Projects for inclusion in this appendix.